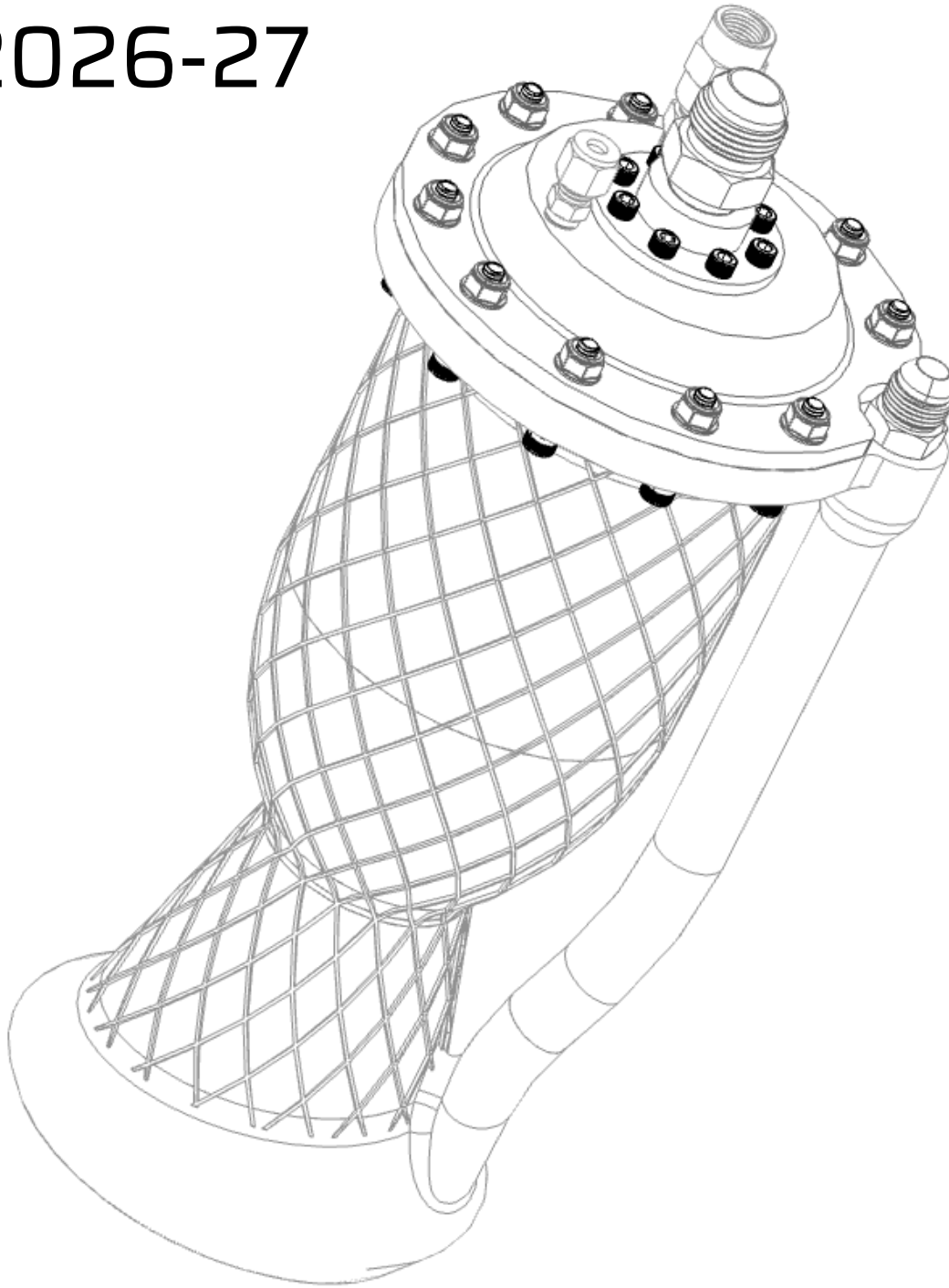


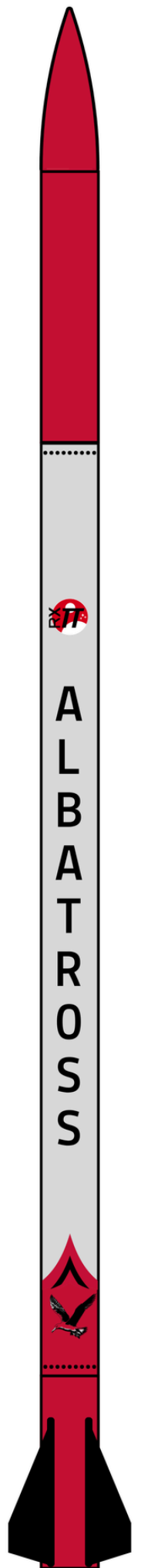


RPI
Experimental
Propulsion Initiative

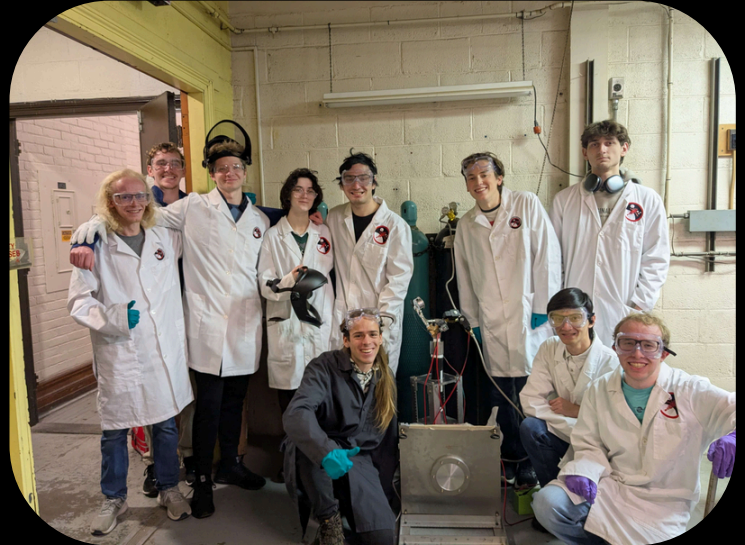
Sponsorship Package 2026-27



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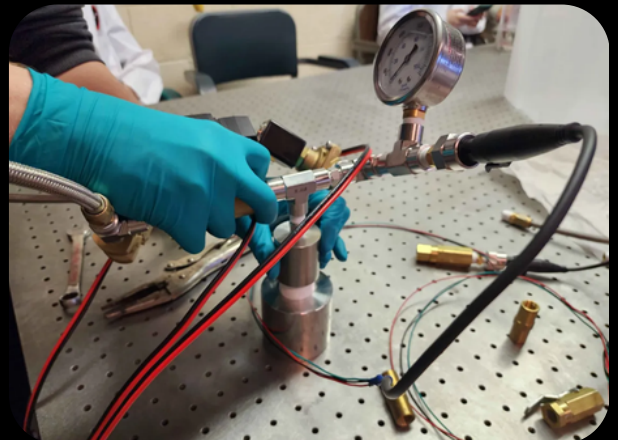


The RPI Experimental Propulsion Initiative (RXPI)



Our team formed under the RPI Spaceflight Society with the goal of developing New York's first student-designed liquid rocket engine. As we conclude a successful cold-flow testing campaign and move toward hotfire, we are in the development stage for our first liquid flight vehicle.

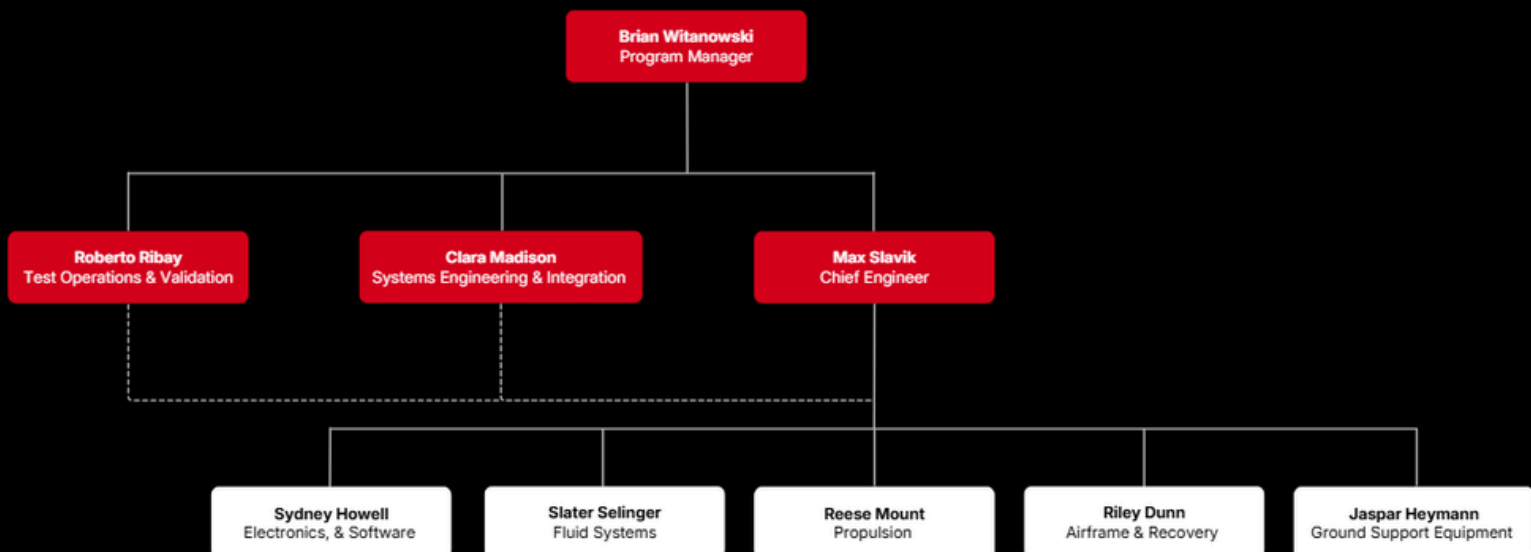
RXPI operates under an industry caliber systems engineering program, prioritizing member involvement, safety, documentation, and efficiency.



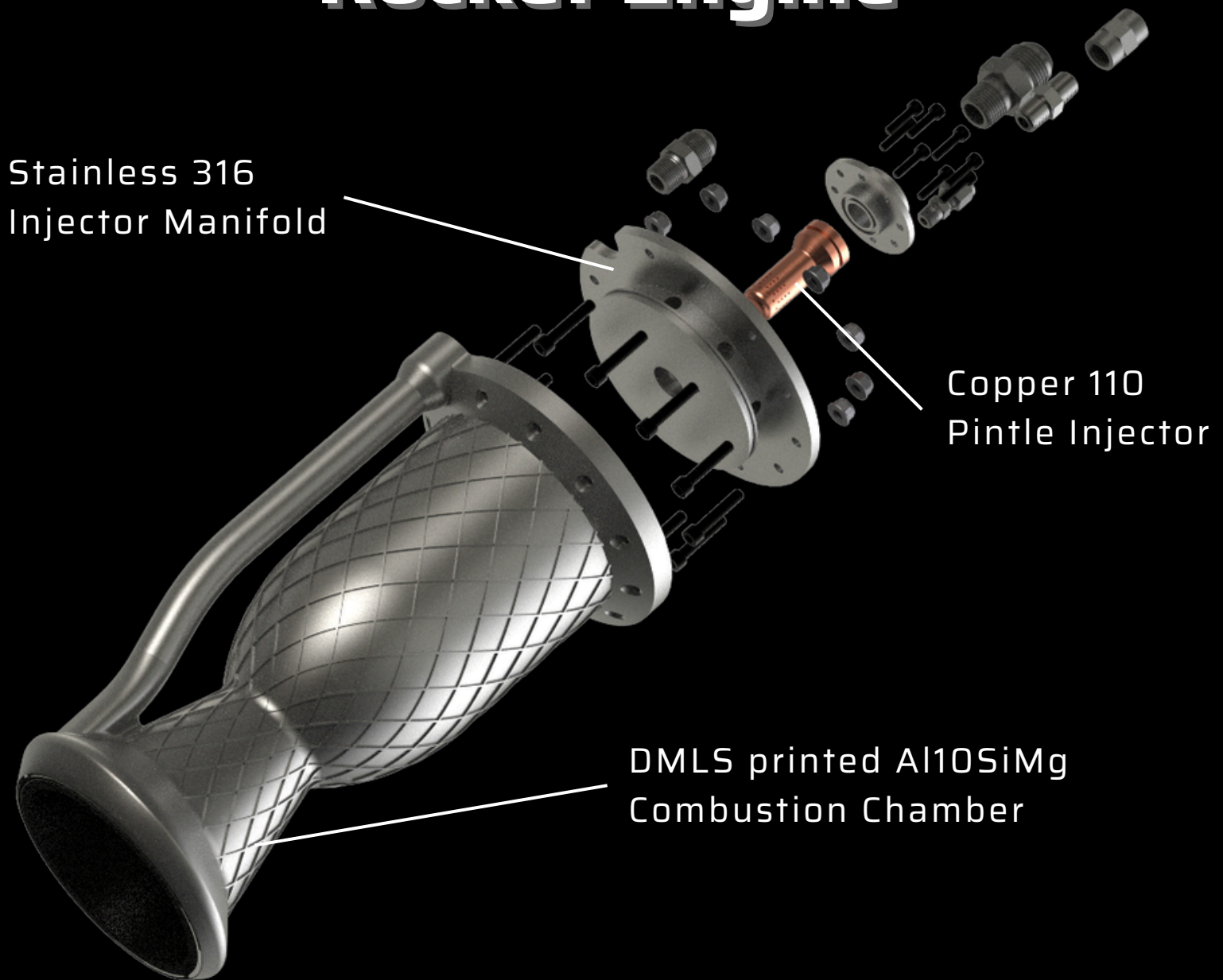
Team Structure

AIRFRAME 8 AIRFRAME & RECOVERY	ES 8 ELECTRONICS & SOFTWARE	FLUIDS 8	PROPULSION 6	GSE 3 GROUND SUPPORT EQUIPMENT
Alexandre Milbrink	Dylan McGuire	Steven Berry	Arthur Bentley	Jordan Woodman
Tyler Kane	Elliot Bieneman	Braja Allin	Ethan Reid	Brendan Garrigan
Zinnia Hooper	Cole Owings	Anthony Khadinovich	Zach Dittus	Colin Pitta
Hunter Szkodzinski	Nicholas Sharp	Ahmed	Aidan Morrison	
Megan Santanen	Andrew Huang	Zen Pinkenburg	Adan Montolio	
Kaitlyn Fleming	Lee Lin	Evan Martynuik	Sara Nickerson	
Jason Treakle	Luca Prattico	Obei Anosike		
Alvin Sheng	Nikolas Daynard	Daniel He		

RXPI utilizes a systems engineering mindset to push large-scale, complex projects forward. Each active team member is responsible for the design, implementation, and testing of a flight vehicle and/or ground support component. This allows every project to contribute greatly to member training and engineering experience.



RPU-2 'Altair' Liquid Rocket Engine



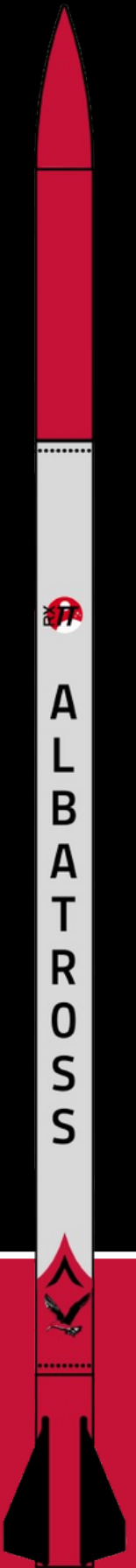
Altair is a next generation propulsion system utilizing additive manufacturing techniques to achieve a complex, optimized geometry. RXPI has developed a custom regenerative cooling solver to verify this complex design, along with a multiphase flow tool to analyze off-design performance.

FV01 'Albatross'

Flight Vehicle

- Target Altitude: 60,000 feet AGL**
- Mach 2.08**
- Thrust: 7kN**
- Lightweight Dual-Acting Vapor Pressurization**
- Regenerative Cooling**
- Multi-function Ground-Support Stand**
- Custom PCB Control Hardware**

Albatross is designed to push the boundaries of collegiate liquid rocketry. Due to its mass-efficient concentric tanks and a regeneratively cooled propulsion system, a world record-breaking altitude will be achieved.



Project Budget

Budgetary Item	Est. Cost (USD)
Feed System (Concentric Tanks, Custom Valves, Hardline Tubing)	\$3,570
Propulsion System (Thrust Chamber Assembly, Injector Components)	\$2,300
Ground Support / Testing Equipment (Test Stand, Remote Fill)	\$4,700
Data Acquisition, Sensors, and Custom Electronics	\$1,295
Recovery Avionics & Hardware, Aerostructure, and Subscale Testing	\$2,850
Tools, Consumables, and Propellant Costs	\$2,400
FAR Launch Fee, Travel Fees, Camping, and Truck Rentals	\$3,700
Total Project Expenditures	\$20,815

Help Our Team Build The Future of Collegiate Liquid Rocketry!

Sponsorship Tiers

INCONEL - Industry Partnership

Donation OR Material Sponsorship over \$5000 USD

- Name prominently listed on test stand during static fire
- Name/Logo highly visible on Flight Vehicle
- Invitation to Design Reviews
- Direct Recruiting Access
- Spotlight posts on our Social Media

TITANIUM - Component Sponsorship

Donation OR Material Sponsorship between \$1000 and \$5000 USD

- Name/Logo listed on test stand during static fire
- Name/Logo on Flight Vehicle
- Sponsorship post on our LinkedIn Page

SCARLET - Alumni and Small Business Support

Donation between \$500 and \$1000 USD.

- Name/Logo listed on test stand during static fire
- Sponsorship post on our LinkedIn Page